TEAM TECH HEAD

HEALTHCARE TECHNOLOGY · EMERGENCY COORDINATION · HOSPITAL MANAGEMENT

EMERGENCY BED COORDINATION

CareRoute

Smarter Emergency Bed Coordination



Emergency coordination, visualised: requests, compatible beds, verified priority.

THE PROBLEM

During medical emergencies, patients and caregivers may struggle to identify suitable hospitals with available beds and the appropriate level of care. Fragmented information about availability, capacity and emergency priority makes coordination difficult.

WHAT WE ARE SOLVING

Coordinating emergency bed requests with suitable hospital capacity, care requirements and verified priority.

OUR SOLUTION

A structured workflow: emergency requests, compatible hospital recommendations, operator verification, reservations, arrival tracking.

3 role-based workspaces

5 seeded demo hospitals

3 bed states tracked

⚠ Prototype on seeded demo data — not a live hospital feed

TEAM TECH HEAD — TJ Tanay Jorihar

PV Puneet Kumar Verma

SS Shraddha Shetty

LB Lalit Bist

When Every Minute Matters, Coordination Matters

The coordination gaps CareRoute is designed to close.

Hard to find the right bed

Patients may struggle to identify hospitals with suitable available beds.

Care levels differ

General beds, ICU beds and ventilator needs require different levels of care.

Capacity must count

Availability and capacity must be considered when recommending a hospital.

Requests compete

Multiple emergency requests can compete for the same limited capacity.

Unverified severity

Unverified severity claims can complicate fair prioritization.

Confusing journeys

Uncoordinated request handling can make the patient journey confusing.

Patient in emergency → Hospital search → Priority & capacity check → Bed confirmation

CORE INSIGHT

Finding an available bed is not enough. The bed must be compatible, available, and allo through a controlled workflow.

One Coordinated Workflow for Emergency Bed Requests

From the first request to arrival confirmation — one structured, role-based flow.

1 — Submit request
Patient or requester submits an emergency request.

2 — Request details
Patient ID, descriptive location, severity, required care type.

3 — Evaluate fit
Compatibility, availability, distance and capacity.

4 — Verify severity
Hospital operator reviews and verifies priority.

5 — Allocate bed
Temporary hold, reservation, rejection or release.

6 — Track & arrive
Arrival confirmed; request tracked in the workspace.

Patient workspace

- Submit an emergency request
- Request a suitable hospital
- Track the request through its states

Hospital operator workspace

- Review incoming requests
- Verify severity
- Accept or reject requests
- Manage reservation & arrival states

Administrator workspace

- Manage demo requests & management functions
- View emergency records
- Reset demo emergency data

Role-based by design — each role gets its own workspace and only the actions it is permitted to perform.

Beyond Availability: Priority-Aware Bed Coordination

Six mechanisms implemented in the prototype — three carry the core innovation.

Compatibility Matching

Requests carry a required care type; matching respects the level of care the patient actually needs.

General · ICU · Ventilator

Verified Priority

Critical, Urgent and Stable/unverified tiers, FIFO ordering for ties. Self-reported Critical alone does not raise priority.

Critical · Urgent · Stable / unverified

Protected Reservations

Temporary holds carry a five-minute expiry; confirmed reservations and occupied beds are protected from reclamation.

5-min expiry · Confirmed = protected

Temporary bed holds

Holds and reservations expire after five minutes, evaluated on server reads and actions. Occupied beds never expire.

Priority-aware reclamation

A higher verified priority can reclaim a lower-priority temporary hold; confirmed reservations and occupied beds stay protected.

Controlled states & roles

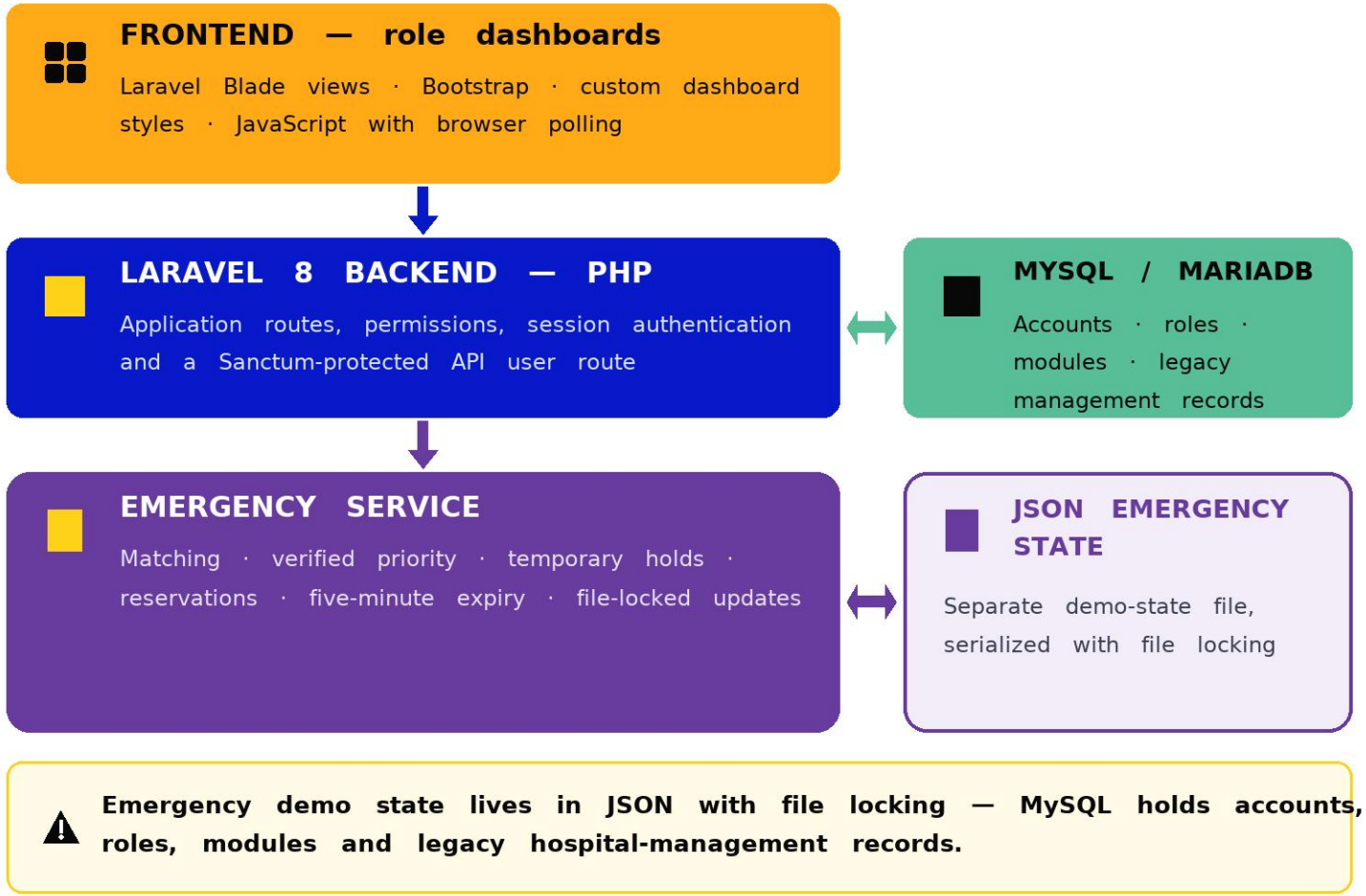
Temporary Hold, Reserved and Occupied are distinct states; patients, operators and admins act only within their permissions.

OUR INNOVATION

Compatible-care matching + operator-verified priority + controlled bed allocation in one workflow

How We Built CareRoute

Laravel 8 + PHP over MySQL/MariaDB, with emergency state kept in file-locked JSON.



⚠ **Emergency demo state lives in JSON with file locking — MySQL holds accounts, roles, modules and legacy hospital-management records.**

TECHNOLOGY STACK

BACKEND	Laravel 8 · PHP
DATABASE	MySQL / MariaDB
EMERGENCY STATE	JSON state · file locking
FRONTEND	Blade · Bootstrap · custom styles · JS polling
AUTH	Laravel UI sessions · Sanctum API route
BUILD TOOLS	Composer · Laravel Mix 6 · Sass · Axios

Five demo hospitals with predefined distances seed the prototype.

From Emergency Request to Occupied Bed

The documented six-step demonstration, run on seeded demo data.

STEP 1

Admin reset

Administrator resets the emergency demo state.

STEP 2

First emergency

Critical / ICU request from an accident location; patient requests Hospital C.

STEP 3

Verify & reserve

Hospital C verifies severity and accepts the request → Reserved.

STEP 4

Second emergency

Second ICU request; Hospital C's reservation is protected, so Hospital B is recommended.

STEP 5

Second reservation

Hospital B verifies and accepts the second request.

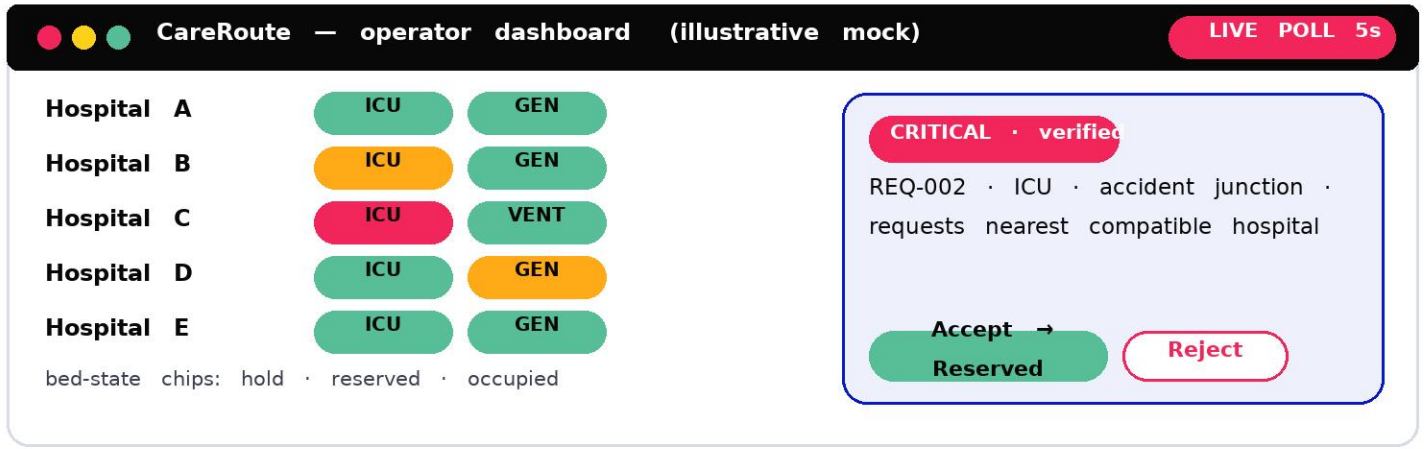
STEP 6

Arrival confirmed

Hospital C marks the first patient arrived → bed state becomes Occupied.

BED STATES Temporary Hold → Reserved → Occupied

also possible: Rejected · Expired (5 min)



DEMO SCOPE

- Five seeded demo hospitals
- Predefined distances between locations
- Admin-resettable emergency state
- Protected reservations steer recommendations

Seeded prototype — not a live connection to real hospital capacity.

Controlled Access. Structured Decisions.

Protections and coordination mechanisms present in the prototype today.

Patient

- Submit & track own emergency requests
- Patient ownership checks enforced
- **Cannot act on other patients' requests**

Hospital Operator

- Review, verify severity, accept / reject
- Manage reservation & arrival states
- **Scoped to the assigned hospital only**

Administrator

- View emergency records
- Reset demo emergency data
- **Emergency reset is admin-only**

PERMISSION BOUNDARY — every role sees and does only what it is allowed

SECURITY & RELIABILITY CHECKLIST

- Session authentication
- CSRF protection on emergency actions
- Patient ownership checks
- Hospital-scoped operator actions
- Admin-only emergency reset
- Decision history for emergency actions
- File locking serializes JSON updates
- Dashboards poll every 5 seconds

PROTOTYPE LIMITATIONS — STATED PLAINLY

Validation, authorization, legacy endpoints, uploads, password-reset workflows and production security require additional review. The README does not establish that the application is production-ready.

Why it matters: users reach only their permitted requests · operators manage only their assigned scope · emergency actions flow controlled routes · concurrent updates are coordinated.

Turning CareRoute into a Healthcare Coordination Platform

A proposed business model for the platform — not an existing revenue stream.

Hospitals, groups & networks

private · multi-hospital · healthcare networks · emergency coordination orgs

→

CareRoute platform

coordination layer: matching, verified priority, reservations

→

Coordinated emergency workflow

structured requests · visible status · protected allocations

POTENTIAL REVENUE STREAMS (PROPOSED)

1 SaaS subscription

Monthly or annual subscriptions for participating hospitals.

2 Enterprise plans

Priced by facilities, user accounts, operational needs and support.

3 Setup & integration

Onboarding, configuration, training and approved integrations.

4 Reporting & analytics

Optional operational reporting and capacity-visibility services.

VALUE PROPOSITION

- Structured emergency request handling
- More visible reservation status
- Better coordination across participating facilities
- Potential reduction in manual coordination effort*
- Foundation for integration with verified hospital systems

TARGET CUSTOMERS

Private hospitals · multi-hospital groups · healthcare networks · emergency coordination organisations · public-sector programs**

GROWTH PATH — PILOT FIRST

Controlled pilot → measure processing time, recommendation accuracy, reservation conflicts, operator workload & adoption → then expand.

*Subject to validation. **Public-sector participation subject to authorization and procurement requirements. No market size, customers or revenue are claimed.

From Prototype to Real-World Healthcare Coordination

Three phases — each clearly separated from what exists today.

PHASE 1

Strengthen the prototype

- Improve validation & automated tests
- Review authorization & security
- Replace fragile demo-state assumptions with robust persistence
- Improve auditability & operational error handling

PHASE 2

Connect real healthcare systems

- Integrate verified hospital bed availability
- Add accurate location & routing services
- Real-time updates via a push-based architecture
- Add reliable notifications
- Data-sharing agreements & access controls

PHASE 3

Intelligent coordination

- Explore demand forecasting
- Investigate discharge prediction
- Evaluate optimization algorithms for matching
- Develop authorized reporting integrations
- Pilot with participating hospitals & measure outcomes

CURRENT PROTOTYPE — TODAY

Seeded hospital data · Five demo hospitals · Fixed distances · JSON emergency state · Browser polling · Rule-based matching & priority

FUTURE ENHANCEMENTS

PLANNED — NOT IMPLEMENTED

Live hospital integrations · GPS-based routing · Real-time push updates · Machine-learning predictions · Authorized government reporting

Every future item is a proposal — nothing above the line is claimed as built.

Better Coordination for Emergency Care

THE PROBLEM

Emergency bed coordination is hard when availability, care needs and priority lack a structured workflow.

OUR SOLUTION

Coordinated requests, compatible hospital recommendations, operator verification, tracked bed-allocation states.

OUR INNOVATION

Compatibility matching · verified priority · temporary holds · protected reservations · role-based actions.

TECHNICAL FOUNDATION

Laravel · PHP · MySQL/MariaDB · JSON emergency state · role-based dashboards.

BUSINESS OPPORTUNITY

A hospital-facing SaaS & integration platform — proposed, subject to validation and adoption.

OUR VISION

Verified, timely information helps providers coordinate emergency capacity more effectively.

“CareRoute — helping the right emergency request reach the right available care facility through smarter coordination.”

TEAM TECH HEAD

TJ Tanay Jorihar

PV Puneet K. Verma

SS Shraddha Shetty

LB Lalit Bist

Thank You

Questions?